

BRIFING NOTE

Knowledge Systems Innovation Project Recommendations

KEY RECOMMENDATIONS FOR ENABLING INVESTMENT IN SCIENCE TECHNOLOGY AND INNOVATION IN KENYA – INSIGHTS FROM THE KNOWLEDGE SYSTEMS INNOVATION (KSI) PROJECT.

This briefing note presents key findings from the assessment of the Science Technology and Innovation (STI) in Rwanda under the auspices of the project '[Understanding Knowledge Systems and what works to promoted STI in Kenya, Rwanda and Tanzania](#)'. The project is funded by the Department for International Development (DFID) through its East African Research Hub.

This briefing note; a) presents key characteristics of Kenya's STI ecosystem necessary for strengthening knowledge production and use; b) key aspects of investments in STI and their socio-economic outcomes; c) opportunities for informal and innovation activity relevant for the Sustainable Development Goals (SDGs); d) policy provocations/recommendations for increasing investments in STI for sustainable development.

APPROACH

The study used multiple sources of evidence ranging from literature review, case study analysis, economic analysis and in country stakeholder consultations to inform practical approaches to promote investments in STI. The study further piloted a framework "the transformative knowledge systems approach as a guiding tool to help direct present and future investments to contribute to sustainable development.

KEY INSIGHTS

Knowledge production: Kenya is characterized by a range of both state and non-state actors steering the production of knowledge in various STI areas. Knowledge production is organized around formal education and research systems and informal systems such as the Technical Vocational Education Training Systems (TVETS) offering vocational training. A number of Public Research Institutions exists in spurring research and innovation in key sectors as development, agriculture and health. Kenya also enjoys a wide cadre of think tanks, international and regional research centers that contributes to its knowledge pool. This open knowledge production system have demonstrated potential for inclusion encompassing an array of actors and partnerships. Specific partnerships, for instance, the establishment of World Bank (WB) Centers of Excellence (CoE) have continued to promote applied research and training relevant to development priorities in local contexts. Other partnership types with development partners have aimed towards building a critical mass of researchers through masters and doctoral training programs. Despite this, brain drain continues to be a great challenge even as governments put measures to mitigate brain drain.

Knowledge impact: Mediation between knowledge, impact and development is still evolving. Innovation support is one of the pathways that the country is pursuing to create knowledge impact. Kenya is known to have a vibrant innovation and entrepreneurship ecosystem through the establishment of a number of both public and private incubators, accelerators and technology hubs. This innovation ecosystem although growing is still immature and stands to benefit from greater support and strengthening. Growing but overlooked is the very significant role of the informal sector as a location of knowledge production, dissemination and impact. The informal sector has been a key actor in recent innovations. It has supported mobile-based finance innovations like *M-Pesa*, *Airtel Money* to thrive including mobile apps like *iCow* and *M-Farm* designed to support small-scale farmers and the development of community cryptocurrencies in

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slums. However, there is very little support in public policy for strengthening formal-informal sector links. Further, the high cost of registration and high tax burden on small businesses actively discourages informal businesses from formalizing, limiting opportunities like financial access to catapult the sector.

Knowledge interactions and inclusion: Knowledge interactions and inclusion are key elements in the pursuit of achieving STI plans in Kenya. This manifest across formal-informal systems as well as across disciplines. Evidence reveal that informal knowledge sector e.g. the '*Juakali*' sector in Kenya builds on peoples' capabilities (e.g. entrepreneurship skills, needs and aspirations), making them able to serve and empower people better. This informal sectors however needs to be better linked to the formal systems and opportunities, be monitored and recognized as an important part of the knowledge ecosystem. There is need for clear structures to help grow the informal knowledge systems into formal ones, largely missing. Although TVETS are some of the efforts being established country wide, as a means of promoting and formalizing informal sectors, they are still characterized by low levels of education which do not match the formal requirements. To this end, many innovation and useful informal knowledge remain untapped for social and economic gains.

Funding mechanisms for knowledge system: Kenya, has established institutional arrangements through the National Research Fund (NRF) to mobilize and manage resources. The fund supports knowledge production through university research and postgraduate studies. There are other financial partnerships such as the Newton Utafiti Fund, established in 2017 in partnership with the British Government supporting collaborative research and knowledge transfer through capacity building. The African Development Bank and the World Bank continues to finance knowledge related activities through collaborative support of university graduates and ICT innovations respectively. This funding support comes in form of grants/credits. Unlike the WB that mainly focuses on funding universities, UK funding majorly supports research institutes. The health sector for example has been hugely supported by UK funding through the KEMRI-Wellcome trust research center in Kilifi. Other government funds include the youth enterprise development fund, Uwezo fund, women enterprise fund, although some of these funds like the Uwezo fund has been greatly tainted by the scandals and corrupt officials running the funding process. Other funding support types like the Digital Village Revolving Fund loans for Pasha Centers are supporting the innovation and incubation hubs through grants and loans to encourage innovators to innovate. These funding schemes remain uncoordinated and therefore hard to link to impacts.

Measurable components of the Knowledge System that work best/worst to achieve socio-economic outcomes: Kenya compares very well in relation to other Low Middle Income Countries (LMICs) on measurable components of the knowledge systems and their association with socio-economic outcomes. It is in a strong position in government support, regulation and governance, and technological capabilities aspects. Mapping research investment across topics/areas from 1990-2017 through publications, reveal research specialization on topics related to health and agricultural sciences. There seems to exist an absence of research capacity in engineering and physical sciences. We confirm a strong dependence on research system on external research funding mostly focussed on health research. The major funders are Wellcome Trust, Bill and Melinda Gates Foundation and European Union. It is left to be seen the reasons why funders focus on health (with limited change in SDG3 indicators) and neglect other research areas. As for the alignment between investments in research on SDGs, the country lags more behind, relative to all other countries, in SDG 9 (Industry, Infrastructure and Innovation). Overallly, there is misalignment of

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patterns of publications and SDG agenda skewed towards specific sectors and need for inclusion of other sectors relevant to the national agenda as outlined in the Big Four agenda.

Overall reflections on the policy environment

The policy environment is still largely focused on creating agencies that could support production, coordination and regulation of knowledge, with emphasis around production of knowledge and research skills for development. While the country has succeeded in setting up these institutions nationally, their effectiveness in achieving broader societal goals is less articulated in the documents. There is no clear pathways in policy documents for ensuring knowledge use beyond production. In addition, accountability measures for tracking investments returns in the different sectors is largely missing. No data or platforms within these agencies exists to view current, previous projected investments and progress towards some of the investments. Whereas government agencies are part of the national framework, the inter-agency /sectorial linkages in the coordination of knowledge production are a bit unclear. Most importantly, the role of non-state actors e.g. the private sector is not well integrated into Kenya's institutional framework for ST&I despite the critical role these actors play in operationalizing the ST&I policies.

Key findings from the case studies

Case study 1: Mechanisms for linking academia with industry in different university settings: Industry-university partnership practice in Strathmore University (SU) and Jomo Kenyatta University of Agriculture and Technology (JKUAT).

The case study explored the practice of industry-university partnerships in different settings. The first being a public university (JKUAT) and the second of a private university (SU). This case was selected because the challenge of leveraging university knowledge and expertise into private sector innovation, which is evident in Kenya's knowledge system, as it is in most countries.

Key message: This case study revealed that promotion of and coordination of knowledge production and use require specific intermediary roles like knowledge brokers to link university-industries and their capabilities have to be enhanced and supported over a long period of time.

Case study 2: Utafiti Sera: a platform for research evidence into policy use and uptake.

This case study explored a novel research into use initiative in Kenya's KS. The case explored *Utafiti Sera* (Kiswahili for research-policy), a platform to get research evidence into policy uptake and use and a dialogue process established by Partnership for African Social and Governance Research. The case was selected because research into use had been identified as a significant challenge in the Kenyan KS. *Utafiti Sera* represents one of a number of recent initiatives in Kenya to build a more productive interface between knowledge producers and users – in this case the social science policy interface. The case study explored the nature of this initiative, its origins and how this has evolved over time, and it's potential to scale to other research into use domains.

Key message: The case study revealed that promotion and coordination of research into use initiatives requires champions to promote research evidence into policy use. There is potential to replicate such models across other SDG areas but with proper coordination and building more consultative platforms for learning and knowledge use.

POLICY RECOMMENDATIONS

Investments in the informal sector

Currently, a number of informal innovations (e.g. the 'Juakali' sector) are emerging and driving Kenya's STI pursuit. While existing policies recognize the contribution of the informal sector, they need go beyond acknowledging the existence of the sector, but also facilitate their transition to formal sectors. This could be done by regularizing the current conditions, providing financial support and putting in place mechanisms through which informal knowledge can be documented and harnessed for greater gains. Creating stronger linkage between informal and formal knowledge systems is likely to contribute to the achievement of ST&I plans. **Recommendation:** *focus new investments on linking the formal and informal sector where specific opportunities exist to align with the SDGs. Also in the process, focus on building and sustaining these capacities and capabilities.*

There is very little support in public policy for strengthening formal-informal sector links. Further, the high cost of registration and high tax burden on small businesses actively discourages informal businesses from formalizing thereby making access to the limited finance available even more difficult. **Recommendation:** *set up a business investment fund specifically targeting informal sector innovation-led business opportunities.*

Learning from various examples of private public partnerships in the STI ecosystem, there is need to maximize the impact of such partnerships. For example our study identified examples where partnership between Kenyatta University and Equity bank established a community outreach and extension program aimed at providing community service opportunities for students. While the initiative has been beneficial to the students and the community at large, high impact could be achieved if training activities could focus on how to manage the fund while putting infrastructure in place to bring about connectivity. **Recommendation:** *investments in the informal sector should make deliberate attempts to understand and invest in the key enablers required to facilitate the informal linkages.*

Currently, we noted recognition of indigenous knowledge in Kenya through the establishment of Resource Centre for Indigenous Knowledge (KENRIK) and the Centre for Biological Diversity (CBD). These institutions are engaged in documenting, preserving and disseminating indigenous knowledge held by different communities across Kenya. **Recommendation:** *indigenous knowledge resource centres are valuable, but pretty isolated. Examination of how to leverage knowledge repositories for actual impact should be undertaken and an assessment of how they can influence research agendas in the main stream.*

Investment in research and data for decision making;

(i) Innovation and incubation hubs: While the innovation ecosystem is currently expanding, there is lack of; a) clear models around the incubation programs and the accelerator programs; b) various target groups; c) nature of support and programmes incubated; d) data on the impacts and success stories of these initiatives. **Recommendation:** *more analytical studies that provides detailed data and information regarding these incubations/innovation hubs to attract possible investments and partnerships.*

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(ii) Tracking investments returns: The study highlighted a number of funding schemes which are currently uncoordinated thereby making it difficult to track investments in STI. **Recommendation:** *invest in strengthening efforts in the existing accountability measures for tracing /tracking investments returns in various sectors. Invest in data or platforms within national agencies where current or previous investments and progress towards some of these investments can be viewed.*

Investments in knowledge exchange platforms

The case study analysis revealed that knowledge exchange platforms, for example the *Utafiti Sera* research into policy use platform, have been very successful in marshalling evidence on different social policy themes and communicating this effectively. **Recommendation:** *investments in knowledge exchange platforms are a potentially powerful way of leveraging impact from research investments. However complimentary investments need to be made in governance structures to ensure inclusiveness of process and direction.*

Investments in knowledge intermediaries/champions and frameworks to support knowledge brokerage

The promotion of and coordination of knowledge production and use require specific intermediary roles like (i) knowledge brokers and (ii) research champions to link university-industries and research evidence into policy use respectively. **Recommendation:** *invest in knowledge brokerage including enhancing their capabilities and support over a long period of time. There is potential to replicate the research into use model of promoting stakeholder conversations to enhance research use across other SDG areas but with proper coordination and building more consultative platforms for learning and knowledge use.*

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